

Lecture #0 Administrative Elements

Fundamental Algorithms

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Core Team

- English track
 - Rodica Potolea
 - Professor, Computer Science Department
 - Rodica.Potolea@cs.utcluj.ro (Room C09)
- Romanian A track
 - Raluca Laura Portase
 - Lecturer, Computer Science Department
 - Raluca.Portase@cs.utcluj.ro (Room M03)
- Romanian B track
 - Camelia Lemnaru
 - Professor, Computer Science Department
 - Camelia.Lemnaru@cs.utcluj.ro (Room M03)
 - Ciprian Oprisa
 - Assoc. Professor, Computer Science Department
 - Ciprian.Oprisa@cs.utcluj.ro (Room M03)
- Core TAs: Richard Ardelean, Vlad Negru, Alex Lapusan

Structure of the course

- Lectures
 - <https://moodle.cs.utcluj.ro/course/view.php?id=775>
 - Slides + discussions + pseudocode
 - Open course with Q&A sessions: ask questions when in doubt!
- Tutorials (Seminars)
 - Problem solving – analysis and design, evaluation, comparisons
 - Pseudocode
- Laboratory sessions
 - Problem solving (alg. implementation, testing and evaluation)
 - C/C++
 - same content for all, each group with a different faculty member or (former) PhD/graduate/master student

Lab sessions info – Romanian Tracks

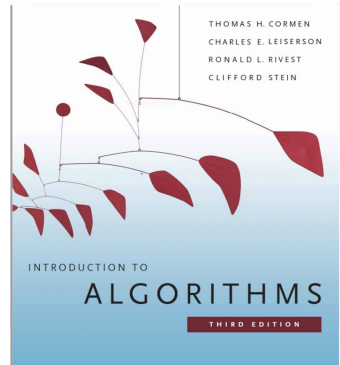
Group	Enrollment Key	TA
30221	Group@30221	Luisa Voicu
30222	Group@30222	Alex Lapusan
30223	Group@30223	Alin Anghelus
30224	Group@30224	Raluca Portase
30225	Group@30225	Alex Lapusan
30226_1	Group@30226_1	Richard Ardelean
30226_2	Group@30226_2	Anca Iordan
30227	Group@30227	Richard Ardelean
30228	Group@30228	Alex Lapusan
30229	Group@30229	Mihai Feier
302210	Group@302210	Richard Ardelean
32721 (CSC)	Group@32721	Anca Iordan

Lab sessions info – English Track

Group	Enrollment Key	TA
30421	Group@30421	Vlad Negru
30422_1	Group@30422_1	Vasile Suciu
30422_2	Group@30422_2	Vlad Negru
30423_1	Group@30423_1	Vasile Suciu
30423_2	Group@30423_2	Vlad Negru
30424	Group@30424	Remus Petrache
30425	Group@30425	Vlad Negru

"The Bible"

- Cormen, Leiserson, Rivest, (Stern), "Introduction to Algorithms", first edition 1990 (second/third edition 2001), fourth edition 2021, MIT Press
- e-copy - moodle
- hard copy - CS Department Library, Baritiu 26-28, Room M04





Evaluation (1)

- Mid-Term Exam (**MTG**)
 - During lecture / separate sessions – (week 6 or 7 - TBD)
 - **MTG** represents 20% of the Final Grade

Evaluation (2)

- Hands on evaluation 1 / weekly laboratory assignments
 - Stay in your group, presence **mandatory**
 - 10 assignments
 - Every (other) lab deadline on an assignment (various thresholds; we encourage evolution & knowledge/skills accumulation)
 - Late assignments policy:
 - Some assignments can be submitted 1 week late: 80% of max grade
 - More than 1 week, no grade (0) on the given assignment
 - Plagiarism policy – **0 tolerance!!! Don't even try!**
 - Must obtain an average grade ≥ 5 (**Ho1G**)
- Hands on evaluation 2 /laboratory exam, week 14
 - Must obtain a grade ≥ 5 (**Ho2G**)
- Lab Grade (**LabG**) = $0.67 * \text{Ho1G} + 0.33 * \text{Ho2G}$
- **LabG** represents 30% of the Final Grade

Evaluation (3)

- Mid term (**MTG**)
 - 20% of the Final Grade
- Hands on evaluation 1&2 (**LabG**)
 - 30% in the Final Grade (need grade of 5 or more to be allowed to take the Final Exam)
- Final Exam (**FEG**)
 - 2-3h examination, open books
 - partly on moodle (for FE grade < 7), optionally oral examination, for FE grade > 7 ; restrictions apply - TBD
 - algorithm traces, questions, algorithm design for specific problems, complexity analysis; open books
 - 50% of the Final Grade

Evaluation (4)

- Mid term (**MTG**)
 - 20% of the Final Grade
- Hands on evaluation 1&2 (**LabG**)
 - 30% in the Final Grade (need grade of 5 or more to be allowed to take the Final Exam)
- Final Exam (**FEG**)
 - 50% of the Final Grade - 35% comprised of the mandatory Moodle examination (**MEG**), and 15% of the optional oral examination (**OralG**)
- bonuses apply for good seminar/course activity
- **FG** = $0.2 * \text{MTG} + 0.3 * \text{LabG} + (0.35 * \text{MEG} + 0.15 * \text{OralG})$
+ **Bonus**
- seminar activity presence is **MANDATORY!** (ECTS guide)

What to expect

- *NOT* a Programming course
- *NOT* a Data Structures course
- *It IS* a course on Fundamental Algorithms, learn (to):
 - *evaluate* algorithm *performance*
 - *compare performance* of different algorithms
 - *design efficient* and *optimal* algorithms
 - *identify a solution* to a given problem
 - *specific efficient algorithms* on *fundamental problems*